

Product Requirements Document and UX Research Report: BharatQuest MVP Golden Path

Executive Overview and Hackathon Constraints

The digital financial landscape in India has undergone an unprecedented transformation, with the Unified Payments Interface (UPI) processing over fifteen billion transactions monthly and rural internet user bases expanding by tremendous margins. However, this rapid onboarding of first-time digital participants in "Bharat"—the vast, non-urban demographic of India—has created profound vulnerabilities. Unaccustomed to the deceptive visual and psychological patterns employed by malicious actors, these users frequently fall victim to phishing, fake reward links, and social engineering attacks. To address this critical gap in digital financial literacy, the proposed product, "BharatQuest," is conceptualized as a voice-first, offline-capable financial survival simulator. Operating fundamentally differently from traditional educational applications that rely on static text and passive reading, BharatQuest integrates the experiential, trial-and-error simulation mechanics popularized by the Banzai financial literacy platform with official scam-awareness curricula developed by the Reserve Bank of India (RBI).

The purpose of this comprehensive Product Requirements Document is to define the exact user interface (UI) and user experience (UX) specifications for the Minimum Viable Product (MVP) of BharatQuest, specifically tailored for a highly constrained three-day hackathon environment. The fundamental constraint of this hackathon necessitates the abandonment of fully dynamic game engines, randomized event generation, and complex economic variables. Instead, the engineering and design focus must be channeled entirely into a single, perfectly executed, linear demonstration scenario known as the "Golden Path." This interactive demo is explicitly engineered to pitch the core behavioral learning loop to the hackathon judging panel.

The Golden Path demo mandates a rigid, six-state sequence: an onboarding dashboard featuring hardcoded starting metrics, an event trigger simulating a fake UPI reward notification, a conversational interface with a manipulative scammer non-player character (NPC), a binary decision gateway, a branching consequence engine demonstrating immediate visual feedback, and a concluding simulated cloud synchronization animation. This document provides exhaustive, pixel-level requirements for each of these states, detailing the psychological rationale, the vernacular design principles, and the specific rendering methodologies required to execute this sequence within the strict temporal limits of the hackathon.

Foundational UX/UI Architecture for the Bharat Demographic

Designing digital interfaces for the rural Indian demographic requires a radical departure from Western, urban-centric design paradigms. The target users often operate within severe constraints, including low digital literacy, variable and unstable network connectivity, and a strong preference for auditory and visual communication over text-heavy interfaces. Consequently, the UI/UX

framework for the BharatQuest MVP integrates several foundational principles to optimize engagement, ensure comprehension, and foster digital trust.

The interface prioritizes a visual-first and icon-based design language. Ethnographic research into rural application adoption indicates that users respond significantly better to images, icons, and graphics that resonate with their cultural and contextual familiarity, which actively reduces cognitive load and allows for navigation via visual recognition rather than reading. In the context of BharatQuest, this translates to utilizing highly recognizable agrarian and financial imagery—such as skeuomorphic currency notes, farm assets, and official-looking badges—to represent abstract financial concepts. Furthermore, the interface must embody the principle of "digital dignity," ensuring that the UX never makes the user feel uneducated or intimidated by complex navigation architectures. To accommodate the low attention spans and high cognitive friction often experienced by users multitasking in rural environments, the Golden Path is structured around micro-flows and single-action screens. Each state in the sequence demands only one primary interaction, completely eliminating nested menus and secondary actions to ensure the judging panel experiences the exact intended narrative without any possibility of deviation.

Equally critical is the implementation of a "voice-first" modality, which serves as the primary conduit for information delivery within the application. While the hackathon constraints limit the dynamic generation of audio, the UI must visually simulate a deeply integrated voice assistant powered by technologies akin to the Bhashini API or Sarvam AI. Voice-first UX in regional languages is not merely a feature; it is a mechanism for removing fear and building emotional connection. The interface must utilize hybrid UX design, where voice cues are paired with localized, conversational text (e.g., using natural, reassuring phrases rather than robotic banking jargon) and prominent visual anchors, such as pulsing microphone icons, to confirm system activity. Finally, the overarching design must incorporate an offline-first philosophy. Recognizing that rural areas frequently experience "Shadow Zones" with limited connectivity, the UI must never present infinite loading spinners or broken interface elements, relying instead on localized storage solutions like WatermelonDB or SQLite, which will be visually simulated in the final state of the demo.

Pedagogical Framework: Integrating Banzai Mechanics and RBI Guidelines

The educational efficacy of the BharatQuest MVP relies on the synthesis of two distinct frameworks: the experiential learning methodology of the Banzai platform and the authoritative regulatory messaging of the Reserve Bank of India. The Banzai curriculum is predicated on the philosophy that students learn best by doing, specifically by navigating real-world financial dilemmas and experiencing the unexpected consequences of their actions within the safety of a simulated environment. Rather than relying on simplified grading or theoretical lectures, Banzai subjects users to trial-and-error scenarios where one-off decisions quickly aggregate into long-term financial impacts. BharatQuest adapts this life-simulation dashboard approach, transplanting the mechanics of managing a budget or a business into the context of a rural village economy. The interface must reflect this by prominently displaying dynamic metrics that react immediately to user inputs, creating a direct cognitive link between behavior and outcome.

Simultaneously, the narrative content and visual authority of the simulation are heavily drawn from the RBI's extensive public awareness campaigns, most notably the "RBI Kehta Hai" (RBI Says)

initiative. The RBI utilizes multi-media and multilingual campaigns to educate citizens on safe digital banking practices, explicitly warning against sharing One-Time Passwords (OTPs), clicking unknown links, and falling prey to fictitious lottery or UPI reward offers. The BharatQuest UI integrates these guidelines not merely as text, but as structural elements of the consequence engine. When the user experiences a failure state within the demo, the resulting educational overlay must mirror the visual authority of the RBI campaigns, utilizing the standardized messaging, recognizable typography, and severe warning tones associated with official regulatory communications to maximize the impact of the simulated lesson.

State 1: Onboarding and the Village Farm Anchor

The initial state of the Golden Path demo serves as the critical anchor point, immediately orienting the hackathon judges to the product's core gamified environment. This onboarding screen bypasses all standard login, authentication, or dynamic character creation flows to adhere to the strict temporal limits of the hackathon presentation. The user is instantly immersed in the regional-language friendly dashboard, which visually establishes their status within the "Village Farm" simulation.

The layout of this dashboard is engineered to draw the user's eye to the most critical economic variables before any disruptive events occur. The upper quadrant of the screen functions as a persistent Heads-Up Display (HUD) containing two explicitly hardcoded starting values: a monetary balance and a behavioral metric. The primary wealth indicator displays a starting "Balance" of exactly ₹5,000. This specific amount is chosen because it represents a tangible, significant, yet realistic sum for a rural demographic, maximizing the psychological impact of potential future loss. Adjacent to the balance is the "Trust Score," hardcoded to a starting value of 80/100. The implementation of a Trust Score is a direct adaptation of Banzai-style life simulation mechanics, providing a quantifiable metric for social reputation and digital hygiene. Starting the user at 80 out of 100 is a deliberate psychological design choice; it informs the user that their standing is currently good but imperfect, triggering an innate psychological drive (loss aversion) to protect and improve this metric.

The central canvas of the screen features a static, vibrant 2D illustration of a rural village farm, providing the necessary cultural and environmental context for the "Bharat" user persona. This imagery acts as the visual representation of the user's current baseline state of well-being. To reinforce the voice-first nature of the application, a prominent, persistent Voice Assistant icon—styled as a localized, friendly "Fortune Teller" motif or a simple, glowing microphone—is anchored to the bottom corner of the interface. Upon the screen rendering, the hardcoded values of ₹5,000 and 80/100 perform a rapid, 1.5-second count-up animation to draw the user's attention, establishing the baseline before the system programmatically initiates the next state after a strict three-second delay.

UI Component Specifications for State 1

Component Name	Visual Architecture	Typography & Color Matrix	Functional Purpose
Balance HUD	Top-left positioning; skeuomorphic stack of rupee	24px Bold Sans-Serif; Text: "₹5,000"; Color:	Establishes baseline virtual wealth; anchors

Component Name	Visual Architecture	Typography & Color Matrix	Functional Purpose
Trust Score HUD	notes icon.	Dark Green (#105C2E).	financial context.
	Top-right positioning; glowing shield icon.	24px Bold Sans-Serif; Text: "80/100"; Color: Golden Yellow (#D4AF37).	Establishes baseline behavioral reputation; triggers loss aversion.
Environment Canvas	Central positioning; vibrant 2D illustration of a healthy village farm.	N/A	Provides cultural resonance and emotional baseline.
Voice Anchor	Bottom-right fixed positioning; circular floating action button with microphone glyph.	N/A	Visually simulates the presence of the Bhashini/Sarvam AI voice engine.

State 2: The Event Trigger and Scam Psychology

Following the brief onboarding period, the narrative momentum of the MVP is forcefully initiated through State 2: The Event Trigger. The core pedagogical thesis of BharatQuest is that financial survival cannot be taught through passive reading; it must be experienced through realistic, high-pressure scenarios. Consequently, this state introduces a deceptive in-game notification that perfectly mimics the mechanics of a real-world digital fraud attempt.

The notification is designed to exploit the vulnerabilities commonly targeted in rural digital adoption. It manifests as a sudden, modal pop-up overlaying the top half of the Village Farm dashboard. The visual presentation leverages "visual trust" by replicating the exact design patterns, green checkmarks, and brand mimicry used by genuine UPI payment applications. The pop-up delivers a fake "UPI Reward" message, a prevalent scam vector where fraudsters trick individuals into believing they are receiving funds, ultimately manipulating them into entering their PIN and authorizing a debit.

The text of the notification is deliberately crafted to induce greed and urgency, reading: "Congratulations! You have won a UPI Reward of ₹2,000. Claim immediately.". To amplify the psychological pressure, the UI incorporates a hardcoded, ticking countdown timer (e.g., starting at 02:59 minutes) displayed in a high-contrast red hue beneath the primary text. Scammers consistently utilize this tactic of temporal urgency to bypass a victim's logical reasoning faculties and force impulsive action. The visual arrival of the notification is tightly synchronized with an auditory cue—a familiar, positive "cash register" or digital ping sound. This auditory stimulus acts as a classical conditioning trigger, automatically associating the pop-up with genuine financial gain in the user's mind. The user is provided no option to dismiss this notification; after a 2.5-second hardcoded delay, the application forces a transition to the conversational interface to escalate the scam.

State 3: The Scammer Interaction and Conversational UX

State 3 shifts the interface paradigm from a dashboard to a direct conversational user experience, introducing the primary antagonist of the demo: the Scammer NPC, operating under the guise of a

"Fortune Teller" or customer service agent. This state is designed to replicate the insidious social engineering tactics deployed through messaging platforms like WhatsApp, which are pervasive in rural India and frequently used as vectors for impersonation and phishing attacks.

The screen layout transitions into a vertically split-pane architecture. The upper portion displays the 2D avatar of the Scammer NPC, designed to project false authority through professional or culturally trusted visual signifiers. The lower portion of the screen renders a simulated chat interface, complete with message bubbles, timestamps, and typing indicators. Adhering to the strict MVP constraints, the chat flow is entirely pre-scripted and automated, requiring no actual text input from the user.

The sequence of messages is timed to maximize psychological pressure. The NPC initiates the interaction with a deceptive greeting, validating the fake UPI reward. Following a brief, simulated "typing" delay, the NPC delivers the critical manipulation: a message injecting extreme urgency and fear, such as claiming the user's account will be blocked if they do not verify their identity immediately to claim the funds. This mirrors the exact language used by fraudsters who pose as bank officials or NPCI representatives. The final automated message presents a large, highly visible chat bubble containing a fraudulent phishing link, demanding that the user click it to process the reward.

Crucially, this state visually demonstrates the "voice-first" capability of BharatQuest. As the phishing link is presented, the Voice Assistant anchor icon pulses urgently, and a simulated regional voice AI audio track plays, actively warning the user: "Caution. Real banks never ask you to click links to receive money". This creates a profound cognitive dissonance for the user—the intense pressure and promise of wealth from the chat interface battling against the protective warning of the voice AI—perfectly setting the stage for the behavioral test in the subsequent state.

State 4: The Decision Gateway and Cognitive Friction

Following the presentation of the phishing link and the AI warning, the automated chat sequence halts, and the application enters State 4: The Decision Gateway. This state is the pivotal interactive moment of the Golden Path demo, forcing the user to process the conflicting information and make a definitive behavioral choice.

A modal slides up from the bottom of the screen, partially obscuring the chat interface, presenting the user with exactly two mutually exclusive interaction buttons: "Claim Reward" (representing the action of clicking the malicious link) and "Report & Block" (representing the action of detecting the scam and protecting the account). The UI design of this gateway relies heavily on the principles of cognitive friction and behavioral economics.

To accommodate the touch target requirements for rural users operating entry-level smartphones, both buttons are rendered massively, spanning the full width of the modal. However, their visual styling is intentionally engineered to subvert standard user interface expectations, thereby simulating the deceptive nature of digital fraud.

Decision Gateway UI Specifications

Button Designation	Label Typography	Background Color Matrix	Visual Affordance and Iconography
Failure Path Button	"Claim Reward" (20px, Bold, White)	Deceptive Blue (#007BFF)	High drop shadow; subtle pulsing animation; "Gift Box" icon.
Success Path Button	"Report & Block" (20px, Bold, White)	Warning Red (#DC3545)	Flat styling; static presence; "Shield with Slash" icon.

The "Claim Reward" button utilizes a bright, standard "confirmation blue"—a color users have been culturally conditioned to associate with safe, positive progression in digital applications. Furthermore, it employs a subtle pulsing animation, utilizing the human eye's natural attraction to movement to tempt the user into a rapid, unthinking click. Conversely, the "Report & Block" button is styled in a harsh, warning red—a color that inherently triggers hesitation and anxiety. This counter-intuitive color mapping forces the user to pause, override their automatic muscle memory, and actively read and comprehend the text labels. The application remains indefinitely locked in this state until the judging panel or demonstration operator physically taps one of the two buttons, triggering the branching consequence engine.

State 5: The Consequence Engine and Visual Feedback

The defining innovation of BharatQuest, adapted from the Banzai curriculum, is its reliance on immediate, tangible consequences rather than post-action theoretical grading. The moment a button is pressed in the Decision Gateway, the UI instantly transitions to State 5, executing one of two rigidly choreographed sequences based on the user's choice.

Branch A: The Failure State ("Claim Reward")

If the user succumbs to the manipulation and taps "Claim Reward," the interface must simulate the devastating realization of a successful financial drain. The entire screen viewport instantly flashes a severe, semi-transparent crimson red for approximately 400 milliseconds. This full-screen visual shock bypasses the need for textual comprehension, utilizing color psychology to immediately signal a critical error to users of any literacy level.

Following the red flash, the UI aggressively snaps focus back to the HUD Header on the Village Farm dashboard. The balance metric actively animates, the text turning red as it counts down from the initial ₹5,000 to ₹3,000, visually deducting the ₹2,000 penalty. Simultaneously, the Trust Score text flashes red and rapidly ticks down by 20 points, dropping from 80/100 to 60/100. The accompanying golden shield icon visually shatters or dulls to grey, providing a permanent visual marker of the lost reputation.

Once the metric deductions finalize, a massive, un-dismissible educational modal overtakes the screen. This overlay directly integrates the visual authority of the official "RBI Kehta Hai" campaigns. The modal features a thick red warning border and displays the official campaign slogan: "Jaankaar Baniye, Satark Rahiye!" (Be Aware, Be Alert) alongside the RBI's mascot, Money Kumar, if available within the asset library. The text explicitly explains the error: detailing that real institutions never ask users to click links or enter PINs to *receive* funds. This text is concurrently read aloud by the simulated voice assistant in a stern, educational tone, ensuring the pedagogical payload is delivered effectively.

Branch B: The Success State ("Report & Block")

Conversely, if the user correctly overrides the deceptive UI and selects "Report & Block," the system triggers a powerful positive reinforcement loop. The screen flashes a vibrant, reassuring emerald green, accompanied by a positive auditory chime to instantly communicate success.

The HUD Header updates to reflect the victory. While the monetary balance remains secure at ₹5,000, it pulses with a brief green glow to emphasize the successful protection of the asset. The Trust Score text highlights in green and animates upward, adding 10 points to reach 90/100. The shield icon expands slightly and glows, providing lasting visual proof of the user's improved digital hygiene.

The sequence culminates in a celebratory modal overlay. This UI element is adorned with vibrant graphics and centers on a large, intricately designed digital badge explicitly labeled "Scam Defender". The accompanying text and synchronized voice-over warmly congratulate the user for correctly identifying the phishing attempt and protecting their simulated village's economy, heavily reinforcing the desired behavior before dismissing the modal.

State 6: The Sync Fake-Out and Lottie Architecture

The final state of the Golden Path MVP serves a specific technical demonstration purpose for the hackathon judges. A core value proposition of BharatQuest is its offline-first architecture, designed to function seamlessly in the frequent network "Shadow Zones" of rural India. To illustrate this capability without requiring the physical disconnection of the demonstration device, the UI executes a simulated, pre-rendered synchronization event.

Upon the dismissal of the consequence modal from State 5, the user is returned to the updated Village Farm dashboard, where a new, prominent button labeled "Sync Progress" has dynamically rendered at the bottom of the interface. When this button is tapped, the interface darkens via a semi-transparent overlay, and a high-fidelity animation sequence commands the center of the screen.

To ensure pristine, vector-based rendering that does not cause frame-drops on the low-end Android devices typical of the target demographic, this sequence is entirely powered by a lightweight Lottie JSON file. The animation strictly adheres to a three-second timeline. It begins by displaying a cloud icon, accompanied by a smooth, circular progress bar that traces the cloud's perimeter, filling from 0% to 100% over the first two seconds. This fluid progress visualization is critical for users in low-bandwidth environments, preventing the anxiety associated with static loading screens.

Upon the progress ring completing its circuit, the cloud graphic rapidly morphs, utilizing an ease-out-back velocity curve to dynamically draw a bold, green success checkmark (#28A745). The resolution of this checkmark signals the successful transmission of the locally stored gameplay data to the simulated cloud backend. Following a brief hold to allow the user to register the success state, the screen fades to black, gracefully concluding the linear Golden Path demonstration.

Conclusion

The BharatQuest MVP, as defined by this strict, six-state Golden Path, successfully distills the complex requirements of a behavioral financial simulator into a concise, highly demonstrable format suitable for a three-day hackathon. By rigidly excluding dynamic economic variables and

randomized open-world events, the engineering and design efforts can be entirely monopolized by the flawless execution of the user interface transitions, the precise integration of the voice-first auditory cues, and the fluid rendering of the Lottie animations.

This meticulous orchestration—moving the user from a stable economic baseline, through the intense psychological manipulation of a simulated phishing attack, into a forced cognitive decision, and culminating in the visceral, immediate feedback of the consequence engine—provides a complete proof-of-concept. It tangibly proves to the judging panel that the fusion of Banzai experiential mechanics with RBI regulatory guidelines, packaged within a vernacular, offline-first UI architecture, holds the potential to fundamentally transform digital financial literacy across rural India.